

Preparation & Ordering

Inputs & Live State

- ▷ **Pods:** CPU, RAM, duration, earliest start, deadline
- ▷ **Nodes:** capacity, idle/peak power, embodied footprint
- ▷ **Forecast:** hourly carbon intensity per region
- ▷ **Residual:** CPU/RAM per node×slot (initially full)

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Build Priority Queue

- ▷ Compute scheduling window $w_i = D_i - d_i$
- ▷ Sort pods by $(w_i \uparrow, \text{CPU} \downarrow, \text{RAM} \downarrow, d_i \downarrow)$
- ▷ Stable ordering keeps runs deterministic

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Enumerate Feasible Windows

- ▷ Scan timeslots between earliest start and deadline
- ▷ Require CPU/RAM headroom for each occupied hour
- ▷ Visit hours in increasing forecasted carbon

Iterate while queue non-empty

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Carbon-Aware Scoring

- ▷ Score $e_{ijt} = \sum_{\tau=0}^{d_i-1} (p_{ij} \text{CI}_{j,t+\tau} + \text{EC}_j/L_j)$
- ▷ Operational term uses shared idle attribution
- ▷ Embodied term amortised per hour; tie-break on slack

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Commit Placement & Update State

- ▷ Select minimum-emission candidate within tolerance ϵ
- ▷ Update residual CPU/RAM on chosen node for pod duration
- ▷ No feasible match $\Rightarrow$ pod re-enters queue next sweep

Results & Artefact

Outputs & Artefacts

- ▷ CSV placements per algorithm + pods.txt metadata
- ▷ Session logs, emission metrics, performance traces
- ▷ Updated state feeding subsequent scheduling sweeps

Unschedulable pods rejoin